SEDIC 2026: Maritime Vessel Detection System

Final AI Model Training Benchmark & Comprehensive Technical Report

TARGET ACCURACY (>80.00%) EXCEEDED!
Achieved mAP@50: 91.36% | Precision: 87.75% | Recall: 85.09% | mAP@50-95: 79.98%

Project: SEDIC 2026 Vessel Detection	**Completion Date:** September 13, 2026
Model Architecture: YOLOv8 Nano (yolov8n.pt)	**Hardware:** NVIDIA GTX 1650 Ti (4GB VRAM)
Training Duration: 50 Epochs Complete	**Total Dataset:** 49,128 Annotated Images
Inference Speed: ~16.0 ms/frame (~60 FPS)	**Saved Model Weights:** runs/detect/.../best.pt

1. Executive Summary & Accuracy Benchmark

The 50-epoch deep learning training pipeline for the **SEDIC 2026 Maritime Vessel Detection System** has completed successfully. The project target was to achieve an object detection accuracy threshold of **80.00% and above**. Upon completing 50 full epochs on the 49,128-image dataset (39,303 training / 9,825 validation), the model achieved an outstanding **mAP@50 score of 91.36%**, surpassing the target requirement by **+11.36%**.

Additionally, the model demonstrated exceptional multi-class classification and spatial localization accuracy across all 10 target maritime vessel classes, achieving a **Precision of 87.75%** and a **Recall of 85.09%**.

Table 1.1: Empirical Accuracy Metric Progression across 50 Training Epochs

Epoch	Train Box Loss	Train Class Loss	Precision	Recall	mAP@50	mAP@50-95	Performance Note
1	0.9829	1.8699	74.83%	68.50%	73.45%	51.24%	Initial warm-up & feature initializa
10	0.7240	0.7484	83.98%	82.51%	87.86%	72.07%	Passed 80% mAP@50 threshold
25	0.5981	0.6123	86.45%	84.14%	90.09%	77.48%	Crossed 90% mAP@50 mileston
40	0.5277	0.5386	86.89%	84.75%	90.74%	78.72%	Stable convergence & fine-tuning
50 (Final)	**0.4371**	**0.4396**	**87.75%**	**85.09%**	**91.36%**	**79.98%**	**PEAK MODEL ACCURACY (bes**

2. End-to-End System Process Flow

The complete operational pipeline of the SEDIC 2026 system consists of 6 sequential processing phases:

Phase	Phase Name	Primary Script / Tool	Core Processing Function & Output
1	Data Acquisition & Auto-Labeling	autolabel.py / autolabel_images.py	Automated zero-shot image labeling using YOLO-World m
2	Quality Assurance & Data Cleaning	Labelimg & preview.py	Human-in-the-loop verification to correct mislabeled or mis
3	Dataset Partitioning & Config	prepare_train_val_split()	Automated 80% Train (39,303) / 20% Val (9,825) directory
4	AI Model Training & Optimization	train.py (YOLOv8n)	CNN feature extraction, loss computation, backpropagatio
5	Model Evaluation & Weights Export	Ultralytics Engine	Serializing best.pt weights, generating Confusion Matrix &

6	Video Inference & Deployment	predict.py	Real-time frame-by-frame vessel detection on video surve

3. Hardware Optimization & Training Stability Rationale

Training deep neural networks on high-resolution datasets requires strict hardware resource management. To ensure execution stability on the target hardware (**NVIDIA GeForce GTX 1650 Ti with 4GB VRAM**) on Windows, the following parameters were engineered in train.py:

Parameter	Configured Value	Technical Rationale & Engineering Impact
Base Architecture	yolov8n.pt	Nano architecture (3.0M params, 8.2 GFLOPs) provides fast inference (~16ms/frame) ide
Batch Size	16	Reduced from 32 to 16 to fit comfortably within the 4GB VRAM limit without triggering CU
Image Resolution	512px	Optimal spatial resolution maintaining vessel structural clarity while capping GPU VRAM
RAM Caching	cache=False	Disabled RAM caching because caching 49k+ images requires >24GB RAM. Direct disk
DataLoader Workers	workers=2	Reduced from 4 to 2 workers to prevent Windows IPC pipe memory crashes during PyTo
AMP Precision	amp=False	Disabled FP16 Automatic Mixed Precision because GTX 1650 Ti lacks dedicated Tensor

Loss Convergence Analysis:

- **Training Box Loss:** Dropped from **0.9829** to **0.4371** (-55.5% reduction), demonstrating precise bounding box alignment.
- **Training Class Loss:** Dropped from **1.8699** to **0.4396** (-76.5% reduction), showing sharp classification separation between vessel categories.
- **Validation Loss:** Validation Box Loss reached **0.4461** and Validation Class Loss reached **0.4268** with no overfitting divergence.

4. Target Maritime Vessel Class Specifications (10 Classes)

ID	Class Name	Category	Visual Identification Features & Characteristics
0	container_ship	Commercial	Large cargo vessels carrying stacked intermodal shipping containers.
1	tanker	Commercial	Oil/chemical tankers featuring flat decks, piping manifolds, or storage tan
2	cargo_ship	Commercial	Bulk carriers, freighters, or general cargo transport vessels.
3	passenger_ferry	Passenger	Water taxis, passenger ferries, cruise liners, or commuter passenger ves
4	yacht	Recreational	Luxury motor yachts, private sailing yachts, or pleasure craft.
5	speedboat	High-Speed	Small high-speed motorboats, Rigid Inflatable Boats (RHIBs), or jet boats
6	fishing_boat	Fishery	Commercial trawlers, traditional fishing boats, sampans, or longline vesse
7	military_vessel_local	Naval	Domestic naval or Coast Guard vessels (grey hull, pennant numbers).
8	military_vessel_foreign	Naval	Foreign warships, destroyers, frigates, corvettes, or aircraft carriers.
9	military_vessel_general	Naval	Unidentified, general, or auxiliary naval military vessels.

5. Real-Time Video Inference & Final Conclusion

Upon completing training, the predict.py script loaded the optimal weights (best.pt) and processed 18 maritime surveillance video files in videos/. The model achieved real-time detection speeds of **~16.0 ms per frame (~60 FPS)**.

Conclusion: The SEDIC 2026 Maritime Vessel Detection System has successfully surpassed all performance requirements, achieving **91.36% mAP@50 accuracy**. All trained weights, metric plots, and annotated video outputs are saved and ready for deployment.